

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTCVC303 - BUILDING CONSTRUCTION & DRAWING

Time : 4 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1	Which stone is used to bound facing & backing of wall in stone masonry?
	a. Header
	b. Through
	c. Quoin
	d. queen closer
	Which bond consists of alternate course of header and stretcher.
	a. English
	b. Flemish
	c. single flemish
	d. double flemish
3	Nominal Mix of M15 concrete is ____
	a. 1: 2:4
	b. 1:3:6
	c. 1:1.5:3
	d. 1:1:2
	In column reinforcement main bars & ____ are used.
	a. Links
	b. Stirrups
	c. Distribution
	d. steel bar
5	Diagonal member of battened, ledged & braced door is ____
	a. bracing
	b. batten
	c. ledge
	d. bracket

6	In which type of window, shutter moves one above the other vertically?
	a. pivoted
	b. double hung
	c. sliding
	d. fixed

7	In lean to roof rafters are supported by ____
	a. Rafters
	b. Purlins
	c. beam
	d. wall plate

	There are 3 types of timber floor 1) single 2) double 3) ____
	a. framed
	b. triple
	c. upper
	d. jack arch

9	The bricks should not absorb water more than ____ of their weight, when immersed in water for twenty-four hour
	a. 10%
	b. 15%
	c. 20%
	d. 5%

	Water cement ration of M15 by weight is ____
	a. 0.67
	b. 0.58
	c. 1.00
	d. 0.75

11	In any staircase maximum steps in one flight should not be more than ____
	a. 15
	b. 8
	c. 10
	d. 12

	In which truss straining sill is provided above the beam?
	a. couple
	b. Collar
	c. King post
	d. Queen post

Q2. Solve the following.

- | | |
|---|---|
| A) Explain principles to be observed in brick masonry construction? | 6 |
| B) Analyze single Flemish & double Flemish bond with sketch? | 6 |

Q3. Solve the following.

- | | |
|---|---|
| A) Explain Reinforcement details of column with sketch? | 6 |
| B) Write a short note on ingredient of concrete? | 6 |

- Q4. Solve Any One of the following.
- A) Classify types of doors (any two) & types of windows (any two) with sketch? 12
- B) Explain timber & steel staircase with sketch? 12
- Q5. Solve Any One of the following.
- A) Draw sketch of queen post truss & king post truss ? 12
- B) Describe any four types of flooring ? 12
- Q6. Solve Any One of the following.
- A) What are the advantages & disadvantages of prefabricated structure? 12
- B) Explain advantages & disadvantages of pre fabricated structure ? 12

*** End ***